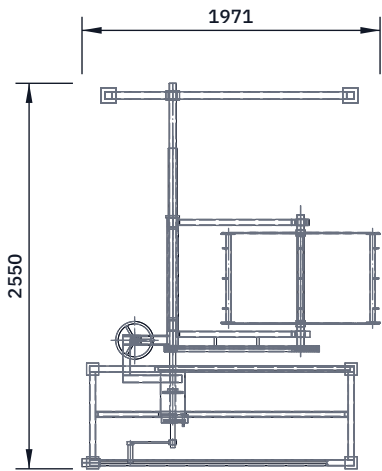


CONCEPT, NOT FOR FABRICATION

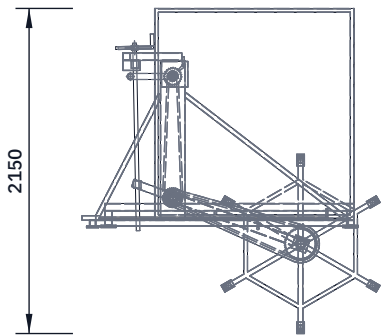
VatReel · github.com/BoujeeEnjinia1701/vatreel

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA from the TRL 3 model (VTR-CAL-001)	2026-10-03	AC
P2	Design for construction (VTR-DDR-002)	2026-10-03	AC



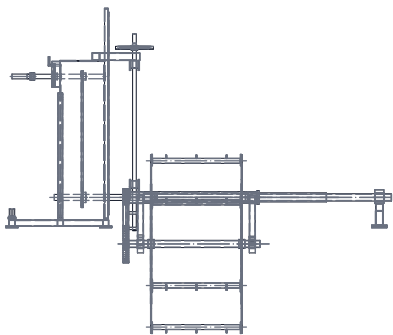
TOP VIEW

Scale 1:50; looking down (along -Z)



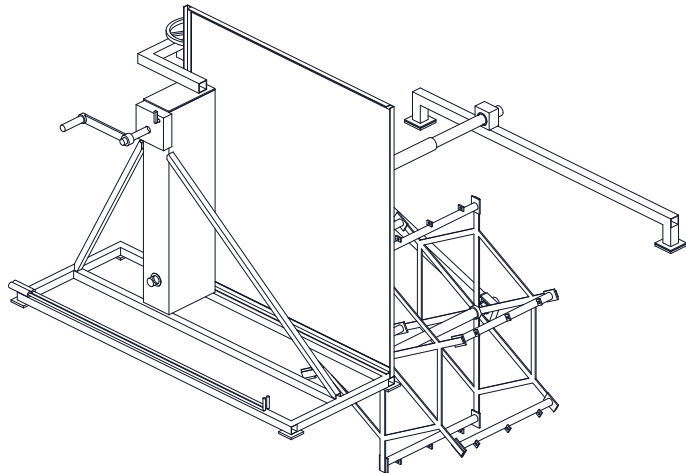
FRONT VIEW

Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; arms 20 deg below level; seen from the front right and above, about 30 deg elevation

KEY DIMENSIONS (MM) AND DATA

Frame 1971 long; 2550 across a 1,600 pit

Pivot line 200 above the rim; arms 900 long

Reel 1180 over spokes, bars at 550 radius

Lowest bar 300 to 1000 below the rim

Lifted: lowest bar 149 above the rim

Pits 1000 to 2,500 wide, 1890 long or more

Crank 1000 high, 280 radius, 6 to 1

About 90 N at the crank, 25 kg on one bar

Lift screw Tr24 x 5, 63 turns; ratchet on crank

Chains 08B: 15/30 and 17/51 teeth

Shield 1320 wide, 60 to 1450 high

About 171 kg; heaviest carry 30 kg by two

PRELIMINARY, NOT FOR FABRICATION

VATREEL

github.com/BoujeeEnjinia1701/vatreel

Hand-cranked reel frame over a pit: general arrangement

DRAWING NO.

VTR-DWG-001

REV

P2

SHEET

1 of 1

SIZE

ANSI B

SCALE

1:50

UNITS

mm

PROJECTION



DATE

2026-10-03

DRAWN

Amish Chadha

MATERIAL / NOTES

304 stainless frame; 316L wetted parts; PTFE and UHMW-PE bushes. See bom/bom.csv

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